# An Introduction to Quantitative Genetics

Comprehensive Module (Detailed + Essentials Versions)

## A Note on the Two Versions in This Document

This document contains **two versions** of the same quantitative genetics module:

1. **The Detailed Version** (approximately 20 pages) – intended for readers who want full explanations, historical context, derivations, examples, glossary, and summary tables.
2. **The Essentials Version** (approximately 4 pages) – intended for readers who only need the core logical flow, key equations, and main takeaways without detailed examples or historical background.

Readers may choose either version depending on their needs. The Essentials version is presented after the Detailed version.

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# DETAILED VERSION

*Approximately 20 pages. Full explanations, historical context, derivations, examples, glossary, and summary tables.*

# 1 Step I: Definition, Necessity, Historical Debate, and Fisher's Resolution

## 1.1 Definition of Quantitative Genetics (QG)

**Quantitative Genetics** is the branch of genetics that deals with **continuously varying traits** (e.g., height, weight, milk yield, growth rate, blood pressure) which are influenced by:

- **Many genes** (polygenic), each with small effect
- **Environmental factors**
- **Interactions** between genes and environment

**Core focus**: Partitioning **phenotypic variance** into genetic and environmental components, and predicting **response to selection** or **resemblance between relatives**.

## 1.2 Why Is Quantitative Genetics Necessary?

Mendelian genetics (single-gene, discrete traits) works beautifully for pea flower color, human blood types, and cystic fibrosis. But many biologically and economically important traits are **not discrete**:

| **Trait** | **Mendelian explanation?** | **Why QG is necessary** |
|---|---|---|
| Human height | No single gene explains it | Hundreds of genes + nutrition |
| Milk yield in cows | No | Many genes + diet, health, management |
| Cholesterol levels | No | Polygenic + diet, exercise |
| Yield in wheat | No | Many genes + soil, rainfall, temperature |

**The necessity**:

- You cannot predict offspring phenotype for these traits using Punnett squares.
- You cannot identify a single "gene for height" that works across populations.
- You need a statistical framework to handle many loci of small effect, environmental noise, and prediction of population means (not individual outcomes).

> **In short**: QG is necessary because most traits in nature and agriculture are continuous, not discrete.

## 1.3 Historical Background: The Mendelian–Biometrician Debate (c. 1890–1918)

| Aspect | Mendelians | Biometricians |
|---|---|---|
| Key figures | Bateson, de Vries, Morgan | Galton, Pearson, Weldon |
| Focus | Discrete, discontinuous variation | Continuous variation, correlations |
| Methods | Controlled crosses, ratios | Statistics, regression, correlation |
| View of evolution | Saltationist (jumps via mutations) | Gradualist (small continuous changes) |
| View of inheritance | Particles (genes) with large effects | Blending (but found regression) |
| Problem | Couldn't explain continuous traits | Couldn't explain discrete inheritance |

**Stalemate**: Both were partially right, but neither could explain the other's evidence.

## 1.4 How Fisher Resolved the Debate (1918)

**R.A. Fisher (1890–1962)**: British statistician and geneticist; founded modern statistics (maximum likelihood, ANOVA). His 1918 paper: *"The Correlation between Relatives on the Supposition of Mendelian Inheritance."*

**Fisher's key insights**:

1. **Many Mendelian loci**: Continuous traits arise from many genes, each following Mendelian rules.
2. **Infinitesimal model**: Each locus has a very small, additive effect. Sum over many loci $\rightarrow$ normal distribution (Central Limit Theorem).
3. **Residual variance**: Accounts for non-genetic factors.
4. **Variance partitioning**: Broke genetic variance into $V_A$ (additive) and $V_D$ (dominance). Showed only $V_A$ transmits reliably.
5. **Predicted correlations**: Derived expected correlations between relatives using only Mendelian principles and allele frequencies.

**The resolution in one sentence**:

> Fisher proved that Mendelian inheritance at many loci, plus residual variance, produces exactly the continuous variation and family correlations that the biometricians had documented.

- **Mendelians were right**: Inheritance is particulate (genes).
- **Biometricians were right**: Continuous traits are heritable and respond to selection.
- **Fisher's synthesis**: Both are true because polygenic traits aggregate many Mendelian loci.

## 1.5 What the 1918 Fisher Paper Established

**A. Quantitative Genetics as a field**: First rigorous mathematical treatment of continuous trait inheritance; provided tools to estimate heritability without knowing individual genes.

**B. Foundations of ANOVA**: Fisher invented ANOVA to partition variance into within-family, between-family, and genetic vs. residual components.

**C. Many statistical ideas originated in Fisher's genetics work**:

- Analysis of variance (ANOVA): partitioning $V_P = V_G + V_E$
- Heritability ($h^2$): $V_A/V_P$ as a predictive parameter
- Regression to the mean: derived from $h^2$ and parent-offspring regression
- Maximum likelihood estimation: first applied to linkage and genetic parameters
- Degrees of freedom: developed for chi-square tests in genetics

**D. Resolution of the debate (lasting impact)**:

- Mendelians accepted that continuous traits are real and heritable.
- Biometricians accepted that inheritance is particulate, not blending.
- Evolutionary synthesis (1930s–40s): Fisher, Haldane, Wright integrated QG with natural selection $\rightarrow$ Neo-Darwinism.

## 1.6 Summary Table: Before and After Fisher

| **Aspect** | **Before Fisher (c. 1910)** | **After Fisher (1918 onward)** |
|---|---|---|
| Continuous traits | Heritable but mechanism unclear | Heritable via many Mendelian loci |
| Discrete traits | Explained well | Still explained well |
| Inheritance mechanism | Disputed (blending vs. particulate) | Particulate (Mendelian) at all loci |
| Prediction of offspring | Regression equations | Regression = $h^2 \times$ midparent |
| Variance | Not systematically partitioned | Partitioned into $V_A, V_D, V_E$ |
| Evolution of continuous traits | Unclear mechanism | Breeder's equation: $R = h^2 S$ |

## 1.7 Key Takeaway for Step I

**Quantitative Genetics is necessary because most heritable traits in nature are continuous, not discrete. The Mendelian–Biometrician debate (1890–1918) was resolved by R.A. Fisher, who showed that many Mendelian loci of small effect, plus residual variance, produce continuous variation. In doing so, Fisher founded QG, invented ANOVA, and provided the statistical framework that underpins modern genetics and evolutionary biology.**

# 2 Step II: What Makes Up the Phenotypic Value of an Individual?

## 2.1 The Core Question

An individual has a **phenotype** ($P$) – a measured trait value (e.g., height in cm, milk yield in liters, flowering time in days).

> **Question**: What determines this phenotypic value?

The answer: Genetic factors and residual factors. But we need to look inside the genetic contribution.

## 2.2 The Additive Basis of Genetic Effects

At every locus that influences a trait, each allele contributes a certain **average effect** on the phenotype.

- Summed over all loci, these average effects give the individual's **additive genetic value**, denoted $A$.
- $A$ is also called the **breeding value** (see Step IV).

> **Key idea**: Additivity means the effects of alleles can be summed. If you know which alleles an individual has, you can add up their average effects to get $A$.

## 2.3 Dominance: Deviation from Additivity Within a Locus

If all loci were perfectly additive, the phenotype would be exactly the sum of allele effects. But often, the combination of *two alleles at the same locus* produces a value that deviates from the additive prediction. This deviation is called **dominance**, denoted $D$.

> **Definition**: $D$ = actual genetic value at a locus minus the additive value predicted from its two alleles.

> **Important**: Dominance exists only when the two alleles at a locus differ. It is a *within-locus interaction*.

## 2.4 Epistasis: Deviation from Additivity Across Loci

Sometimes, the effect of an allele at one locus depends on which alleles are present at *other* loci. This is called **epistasis**, denoted $I$.

> **Definition**: $I$ = actual genetic value across multiple loci minus the sum of the individual locus effects (including their dominance deviations).

> **Important**: Epistasis is a *cross-locus interaction*. It can be extremely complex when many loci are involved.

## 2.5 The Residual Variance ($E$)

In Fisher's formulation, $E$ does **not** simply mean "the physical environment." Rather, it is the **residual variance** – the variation in phenotype that remains after accounting for all identifiable genetic components ($A$, $D$, and $I$).

**What $E$ includes**:

- Physical environmental effects
- Measurement error
- Developmental noise
- G×E interactions (if not explicitly modeled)
- G–E correlations (if not explicitly modeled)
- Any other unexplained sources of variation

**Key point**: $E$ is a **statistical residual**, not a directly measurable biological variable. In a typical quantitative genetics analysis, $E$ is what is left over after fitting genetic components. It is estimated at the population level, not assigned to individuals.

**Historical note**: Fisher's insight was that one could partition phenotypic variance into genetic and residual components **without measuring individual environments**. The residual $E$ could be estimated from family correlations alone. This is why $E$ is defined as "unexplained variance" rather than "physical environment."

## 2.6 The Complete Model

Putting it all together:

$$\boxed{P = A + D + I + E}$$

Where:

- $P$ = Phenotypic value of the individual
- $A$ = Additive genetic value (sum of average effects of alleles)
- $D$ = Dominance deviation (within-locus interaction)
- $I$ = Epistatic deviation (across-locus interaction)
- $E$ = Residual deviation (unexplained variance)

**Important interpretation**: $D$ and $I$ are **deviations from additivity**. If all loci were perfectly additive and independent, $D = 0$ and $I = 0$. In reality, $D$ and $I$ are often present but are usually smaller than $A$ for most quantitative traits.

## 2.7 From Individuals to Populations: The Variance Partition

The model $P = A + D + I + E$ describes an individual. When we consider a population, the variance (spread) of the trait can be partitioned similarly. Under the simplifying assumptions of no G×E interaction and no correlation between $G$ and $E$:

$$\boxed{V_P = V_A + V_D + V_I + V_E}$$

Where:

- $V_P$ = Total phenotypic variance
- $V_A$ = Additive genetic variance (variance of breeding values)
- $V_D$ = Dominance variance (variance due to within-locus interactions)
- $V_I$ = Epistatic variance (variance due to across-locus interactions)
- $V_E$ = Residual variance (unexplained variance, including environment, error, noise)

> **Fisher's complete insight**: The total phenotypic variance in a population can be partitioned into independent components: additive, dominance, epistatic, and residual. Only the additive component ($V_A$) contributes to parent-offspring resemblance and predicts response to selection.

From this, narrow-sense heritability is defined as:

$$h^2 = \frac{V_A}{V_P}$$

Broad-sense heritability (including all genetic components) is:

$$H^2 = \frac{V_A + V_D + V_I}{V_P} = \frac{V_G}{V_P}$$

## 2.8 Returning to the Individual: What Matters for Inheritance?

The variance partition $V_P = V_A + V_D + V_I + V_E$ tells us how variation in the population is structured. But when we ask about **inheritance** – what a parent passes to its offspring – we must return to the individual level.

From the individual-level equation $P = A + D + I + E$, we know that an individual's phenotype is built from four components. However, not all of these components survive the process of reproduction.

> **The key question that drives the rest of this module**: When an individual reproduces, which of these components ($A$, $D$, $I$, or $E$) does it actually pass to its offspring?

The variance partition is essential for understanding the **population-level consequences** of inheritance. But to understand **what is inherited**, we must focus on the individual and the mechanics of meiosis and reproduction.

We will answer this question in **Step III**. First, let us summarize what we have covered in Step II.

### 2.9 Summary of Step II

| Concept | Symbol | Meaning |
|---|---|---|
| Phenotypic value | $P$ | The measured trait value of an individual |
| Additive genetic value | $A$ | Sum of average effects of all alleles (breeding value) |
| Dominance deviation | $D$ | Within-locus interaction (deviation from additivity) |
| Epistatic deviation | $I$ | Across-locus interaction (deviation from additivity) |
| Residual deviation | $E$ | Unexplained variance (includes environment, error, noise) |

**Fundamental equations**:

$$P = A + D + I + E$$

$$V_P = V_A + V_D + V_I + V_E$$

**Take-home message**:

*An individual's phenotype is built from additive effects (each locus contributes), plus dominance (within-locus interactions), plus epistasis (across-locus interactions), plus residual variance (unexplained). Only the additive part will matter for inheritance.*

### 2.10 Historical Note (Footnote for Step II)

**For accuracy**: The model $P = A + D + I + E$ evolved over time.

- **R.A. Fisher (1918)** introduced the partitioning of genetic variance into **additive** ($A$) and **dominance** ($D$) components in his seminal paper "The Correlation between Relatives on the Supposition of Mendelian Inheritance."
- **Epistasis** ($I$) as a formal variance component was developed later, primarily by **Kenneth Mather and colleagues** in the 1940s–1950s, as they extended Fisher's model to include interactions between loci.

For an introductory course, we present the complete model as if all components were always known. The historical detail is provided for completeness but is not essential for understanding the core concepts.

# 3 Step III: What Does an Individual Pass On to Its Progeny?

## 3.1 The Central Question

We have established that $P = A + D + I + E$. Now we ask:

> **When an individual reproduces, which of these components ($A$, $D$, $I$, or $E$) does it pass to its offspring?**

This determines how traits evolve, how breeders improve crops and livestock, and how resemblance between relatives is calculated.

## 3.2 The Short Answer

| Component | Passed to offspring? | Reason |
|---|---|---|
| $A$ (Additive) | **Yes** | Each parent passes one allele per locus. The average effects sum. |
| $D$ (Dominance) | No | Requires a specific *pair* of alleles at the same locus. Meiosis breaks pairs. |
| $I$ (Epistasis) | No | Requires specific combinations of alleles *across loci*. Meiosis scrambles combinations. |
| $E$ (Residual) | No | By definition, not inherited. |

> **The key insight**: Only the **additive component** ($A$) transmits reliably from parent to offspring. This is why $A$ is called the **breeding value**.

## 3.3 Why Dominance ($D$) Is Not Passed

Dominance arises from the *interaction between two alleles at the same locus* – one from the mother, one from the father. When that individual reproduces, it passes only **one** of the two alleles to any given offspring. The specific pair that created the dominance effect is broken.

**Example (one locus)**:

| Genotype | Additive ($A$) | Dominance ($D$) | Total genetic value |
|---|---|---|---|
| $AA$ | +10 | 0 | +10 |
| $Aa$ | 0 | +6 | +6 |
| $aa$ | -10 | 0 | -10 |

An $Aa$ parent has $A = 0$ and $D = +6$. When it reproduces, it passes either $A$ or $a$ (probability 0.5 each). The offspring receives the other allele from the other parent. In no case does the offspring inherit the specific $Aa$ *pair* that gave the parent its $D = +6$. The dominance effect is lost.

> **Conclusion**: Dominance is a *property of a genotype*, not a *property of an allele*. Since parents pass alleles, not genotypes, dominance does not transmit.

### 3.4 Why Epistasis ($I$) Is Not Passed

Epistasis arises from interactions *across different loci*. When an individual reproduces, meiosis and recombination shuffle which combinations of alleles go into each gamete. The specific multi-locus combination that created the epistatic effect is almost always broken.

**Example (two loci)**: Suppose two loci interact: at locus 1, alleles $A$ (+5) and $a$ (-5); at locus 2, alleles $B$ (+3) and $b$ (-3); epistasis: combination $A$ with $B$ gives an extra +4 beyond additive. A parent with genotype $AaBb$ (heterozygous at both) may show epistasis, but gametes receive random combinations ($AB, Ab, aB, ab$). The specific multi-locus combination is broken during meiosis.

> **Conclusion**: Epistasis is a *property of a specific multi-locus combination.* Since meiosis reassorts loci independently, these combinations do not survive intact across generations.

### 3.5 Why the Residual ($E$) Is Not Passed

By definition, $E$ represents unexplained variance – everything not accounted for by genetic components. Physical environment is not encoded in DNA, measurement error is random, and developmental noise is non-heritable. $E$ has no mechanism for inheritance.

### 3.6 Why Additive ($A$) Is Passed

The additive value $A$ is the sum of **average effects of individual alleles**. Each parent passes exactly one allele per locus to each offspring. The average effect of that allele is, on average, the same in the offspring as it was in the parent (assuming the population context is similar). The parent's additive value is the **average** of the additive values it passes to its offspring (across many offspring).

### 3.7 The Crucial Distinction: Genotype vs. Alleles

| Level | What is it? | Passed on? |
|---|---|---|
| Genotype | Specific pair of alleles at a locus (e.g., $Aa$) | No – only one allele per locus goes to each offspring |
| Allele | A single variant at a locus (e.g., $A$ or $a$) | Yes – one per locus per offspring |
| Additive effect | Average contribution of an allele | Yes – transmitted with the allele |
| Dominance | Interaction effect of a *pair* of alleles | No – the pair is broken |
| Epistasis | Interaction effect across *loci* | No – combinations are scrambled |

> **The principle**: Meiosis and reproduction transmit **alleles**, not genotypes, not interactions. Therefore, only effects that can be expressed as sums of individual allele contributions ($A$) are heritable.

### 3.8 Summary of Step III

| Question | Answer |
|---|---|
| What does a parent pass to offspring? | Alleles (one per locus per offspring) |
| Which component of $P$ transmits? | Only the **additive component** $A$ |
| Why not $D$? | Dominance requires a specific *pair* of alleles; the pair is broken in meiosis |
| Why not $I$? | Epistasis requires specific *combinations* across loci; recombination scrambles them |
| Why not $E$? | Residual variance is not encoded in DNA |

**Take-home message**:

> *When an individual reproduces, it passes its alleles, not its genotype. The additive value (A) – the sum of the average effects of those alleles – is the only component of the parental phenotype that reliably predicts the offspring's genetic value. This is why A is called the **breeding value**.*

# 4 Step IV: How Do I Measure What Is Passed On? – Breeding Value

## 4.1 The Problem We Need to Solve

From Step III, only the additive component ($A$) is passed from parent to offspring. But we cannot look at an individual's genotype and simply "add up" average effects because we do not know all the loci involved, their average effects, or the genetic architecture.

**How do we measure an individual's $A$ (its breeding value)?**

## 4.2 The Definition of Breeding Value

**Breeding value** (denoted $A$) is defined operationally as:

> *Twice the deviation of the mean of an individual's progeny from the population mean, when that individual is mated randomly to the population.*

Let:

- $\bar{P}$ = population mean
- $M$ = mean phenotypic value of all progeny produced by individual $i$ (when mated to random individuals from the population)

Then:

$$\text{Breeding value of } i = 2 \times (M - \bar{P})$$

## 4.3 Why Multiply by 2?

The factor of 2 accounts for the fact that the individual contributes only **half** the alleles to each offspring (the other half come from the random mate).

- The difference $M - \bar{P}$ represents the effect of *half* of the individual's alleles (the half that went into the progeny).
- Doubling gives the effect of *all* of the individual's alleles – its full breeding value.

**Important**: This definition assumes random mating with the population.

## 4.4 Understanding Breeding Value Through an Example

**Trait**: Milk yield in cows

| **Quantity** | **Value** |
|---|---|
| Population mean milk yield ($\bar{P}$) | 8000 liters/year |
| Cow A's progeny mean (from random mates) | 8200 liters/year |
| Deviation ($M - \bar{P}$) | +200 liters/year |
| **Cow A's breeding value** | $2 \times 200 = +400$ liters/year |

**Interpretation**: Cow A's alleles, on average, add +400 liters/year to her offspring's milk yield (above the population mean). When mated randomly, her daughters are expected to produce 8200 liters/year on average.

## 4.5 Properties of Breeding Value

**Property 1**: Breeding value is expressed as a deviation from the population mean. The average breeding value of a randomly chosen individual from the population is 0.

**Property 2: Breeding value is not a fixed property of an individual.** Breeding value depends on:

- The **population** (allele frequencies, genetic background)
- The **environment** (if G×E exists)
- The **mating system** (random vs. assortative)

An individual may have a high breeding value in one population but a low breeding value in another.

**Property 3**: The variance of breeding values in a population is the additive genetic variance:

$$\mathrm{Var}(A) = V_A$$

## 4.6 Why Breeding Value Is the Central Concept of QG

| Concept | Role |
|---|---|
| Breeding value ($A$) | What is actually transmitted from parent to offspring |
| Phenotype ($P$) | What we can measure easily, but includes non-transmissible parts ($D, I, E$) |
| Heritability ($h^2$) | The proportion of phenotypic variance due to variation in breeding values: $h^2 = V_A/V_P$ |

**The key insight**: If we know an individual's breeding value, we can predict the average value of its offspring (when mated randomly). If we know the breeding values of a group of selected parents, we can predict the response to selection.

## 4.7 A Practical Challenge: Direct Measurement of Breeding Value Is Difficult

The definition requires mating the individual to many random mates, measuring many offspring, and calculating the mean offspring value. This is time-consuming, expensive, impractical for long-lived species, and impossible if the individual is no longer alive.

**We need an indirect method to estimate breeding value without waiting for progeny data.**

## 4.8 Summary of Step IV

| Concept | Definition |
|---|---|
| Breeding value ($A$) | $2 \times (M - \bar{P})$, where $M$ = mean of progeny from random mates |
| Interpretation | The average effect of an individual's alleles on its offspring |
| Key property | Only $A$ is transmitted from parent to offspring |
| Important nuance | Breeding value is **population-dependent** – it can change if the population or environment changes |
| Variance of $A$ | $\mathrm{Var}(A) = V_A$ (additive genetic variance) |

**Take-home message**:

*Breeding value (A) is the currency of inheritance. It measures what an individual actually passes to its offspring. An individual with breeding value +x will, on average, produce offspring that exceed the population mean by x/2 (because it contributes half the alleles). However, directly measuring breeding value requires progeny data, which is often impractical – so we need indirect methods.*

# 5 Step V: How to Estimate Narrow-Sense Heritability – Mid-Parent Offspring Regression

## 5.1 The Problem Restated

We need a practical, indirect method to estimate:

- An individual's breeding value ($A$)
- The narrow-sense heritability ($h^2 = V_A/V_P$)

  **The solution**: Mid-parent offspring regression.

## 5.2 The Basic Idea

If we measure a trait in parents (mid-parent values = average of two parents) and their offspring, then the **slope** of the regression line estimates $h^2$:

$$\text{Slope} = h^2 = \frac{V_A}{V_P}$$

## 5.3 Why Does the Slope Equal $h^2$?

**Intuitive explanation**:

- The mid-parent value estimates the average breeding value of the two parents.
- Offspring inherit half their alleles from each parent.
- Offspring breeding value = average of parental breeding values (plus random Mendelian sampling).
- Offspring phenotype = offspring breeding value + residual ($D + I + E$).
- Regressing offspring phenotype on mid-parent value captures the proportion of phenotypic variance due to additive genetic effects.

**Key equation (for reference)**:

$$\text{Cov}(P_{\text{offspring}}, P_{\text{mid-parent}}) = \frac{1}{2}V_A$$

$$\text{Var}(P_{\text{mid-parent}}) = \frac{1}{2}V_P \quad \text{(under random mating)}$$

$$\text{Slope} = \frac{\text{Cov}}{\text{Var}} = \frac{\frac{1}{2}V_A}{\frac{1}{2}V_P} = \frac{V_A}{V_P} = h^2$$

**For the introductory student**: The algebra shows that the slope of the mid-parent-offspring regression equals narrow-sense heritability.

## 5.4 A Concrete Example

**Trait**: Plant height (cm)

A researcher measures 100 families. For each family: mid-parent height = average of father and mother; offspring height = average of their progeny.

Regression analysis yields a slope of 0.45. Therefore:

$$h^2 = 0.45$$

**Interpretation**: 45% of the phenotypic variance in this population is due to additive genetic effects. If we select parents with mid-parent value 10 cm above the population mean, we expect offspring to be $0.45 \times 10 = 4.5$ cm above the mean.

### 5.5 Estimating an Individual's Breeding Value from the Regression

Once we have estimated $h^2$:

$$\text{Breeding value} = h^2 \times (P - \bar{P})$$

**Example** (continuing from above): $h^2 = 0.45$, population mean height = 150 cm, individual plant height = 170 cm ($P - \bar{P} = +20$):

$$\text{Breeding value} = 0.45 \times 20 = 9 \text{ cm}$$

**Interpretation**: This plant's alleles are expected to add, on average, +9 cm to its offspring (above the population mean). This is less than its phenotypic deviation (+20 cm) because only the additive part ($h^2$ of the deviation) is transmissible.

### 5.6 Assumptions of Mid-Parent Offspring Regression

For the slope to equal $h^2$, several assumptions must hold:

| **Assumption** | **Why it matters** | **What violates it?** |
|---|---|---|
| Random mating | Affects variance of mid-parent values | Assortative mating |
| No common environment | Offspring should not resemble parents due to shared environment | Parents and offspring raised together |
| No G×E interaction | Genetic effects consistent across environments | Different farms, climates, years |
| Parents are random sample | Should represent the population | Selected parents bias estimate |
| No selection | Parents not chosen based on trait | Artificial selection experiments |
| No sex differences | Heritability equal across sexes unless specific mechanisms | Sex-linkage, maternal effects, imprinting |

> **Important**: In practice, these assumptions are often violated. Heritability estimates must be interpreted with caution.

### 5.7 What If We Use Single Parent Instead of Mid-Parent?

| **Regression** | **Slope equals** |
|---|---|
| Offspring on **mid-parent** | $h^2$ |
| Offspring on **one parent** (e.g., father only) | $\frac{1}{2}h^2$ |

**Why half?** Because one parent contributes only half the offspring's alleles.
**Example**: Regress offspring on father only: slope = 0.225 → $h^2 = 2 \times 0.225 = 0.45$.

> **Practical note**: Using mid-parent gives a direct estimate of $h^2$ and is more precise (averages out measurement error). Separate regressions for each parent can reveal interesting biological phenomena (see Section 5.8 below).

## 5.8 Think About This

**Can the heritability of a given trait estimated via females and via males (i.e., what is on the x-axis in a parent-offspring regression) be different? If so, why?**

**Answer**: Yes, it can be different.
**Possible reasons**:

| **Reason** | **Explanation** | **Example** |
|---|---|---|
| Sex-linkage | Genes on sex chromosomes inherited differently by sons and daughters. | X-linked trait: high heritability father→daughter, zero father→son. |
| Sex-limited expression | Some genes expressed only in one sex. | Milk yield: father has no phenotype, estimated via daughters. |
| Maternal effects | Mother's phenotype directly influences offspring non-genetically. | Mouse body weight: large mother → more milk → large offspring. |
| Parental sex-specific dominance/imprinting | Gene expression depends on parent of origin. | Father's copy silenced, only mother's copy active. |
| Sex-specific environmental effects | Different environments for males and females alter additive variance. | Sexual dimorphism: different nutritional pressures. |

**Practical implication**:

*When estimating heritability, calculate separate regressions for father-offspring and mother-offspring, and for sons versus daughters. If the estimates differ, it tells you something interesting about the genetic architecture (sex-linkage, maternal effects, imprinting, etc.).*

**Default assumption in standard QG**: In the absence of sex-linkage, maternal effects, and imprinting, heritability estimated from fathers and mothers should be equal.

## 5.9 Summary of Step V

| **Concept** | **Explanation** |
|---|---|
| Method | Regress offspring phenotype on mid-parent phenotype |
| Slope | Estimates $h^2 = V_A/V_P$ |
| Why slope = $h^2$ | Covariance = $\frac{1}{2}V_A$; variance of mid-parent = $\frac{1}{2}V_P$ |
| Breeding value prediction | $A = h^2 \times (P - \bar{P})$ |
| Key assumptions | Random mating, no common environment, no G×E, random sample of parents |
| Sex-specific differences | May differ due to sex-linkage, maternal effects, or imprinting |

**Take-home message**:

*Mid-parent offspring regression is the classic method for estimating narrow-sense heritability ($h^2$) and predicting breeding values. The slope tells us what proportion of phenotypic variance is additive genetic – and therefore, how much response we can* *expect from selection. However, the estimate is only valid under specific assumptions. If you estimate heritability separately from fathers and mothers and get different results, that difference tells you something important about the biology of the trait.*

# 6 Step VI: The Breeder's Equation and Predicting Response to Selection

## 6.1 The Question That Drives Quantitative Genetics

We have covered:

- **Step II**: What makes up an individual's phenotype ($P = A + D + I + E$)
- **Step III**: What is passed to offspring (only $A$)
- **Step IV**: How to measure what is passed (breeding value)
- **Step V**: How to estimate heritability ($h^2$) via parent-offspring regression

Now the most practical question:

**If we select the best individuals in a population to be parents, by how much will the next generation improve?**

**The answer**: The **Breeder's Equation**.

## 6.2 The Breeder's Equation

$$\boxed{R = h^2 \times S}$$

Where:

- $R$ = **Response to selection** = mean of offspring  mean of original population
- $h^2$ = **Narrow-sense heritability** = $V_A/V_P$
- $S$ = **Selection differential** = mean of selected parents  mean of original population

## 6.3 Understanding Each Component

**The Selection Differential ($S$)**:

$$S = \bar{P}_{\text{selected}} - \bar{P}_{\text{population}}$$

- Select the very best $\rightarrow S$ large and positive
- Select the very worst $\rightarrow S$ large and negative
- Select randomly $\rightarrow S = 0$

**Example**: Population mean height = 170 cm; selected parents mean = 180 cm $\rightarrow S = 10$ cm.
**The Response ($R$)**:

$$R = \bar{P}_{\text{offspring}} - \bar{P}_{\text{population}}$$

**The Heritability ($h^2$)** acts as a discount factor:

- $h^2 = 1 \rightarrow R = S$ (offspring shift as much as parents)
- $h^2 = 0 \rightarrow R = 0$ (no response regardless of selection)
- $0 < h^2 < 1 \rightarrow R < S$ (offspring shift, but less than parents)

**Why does $h^2$ discount $S$?** Because only the additive part of the parental phenotype is transmitted. Non-additive parts ($D, I, E$) are not inherited.

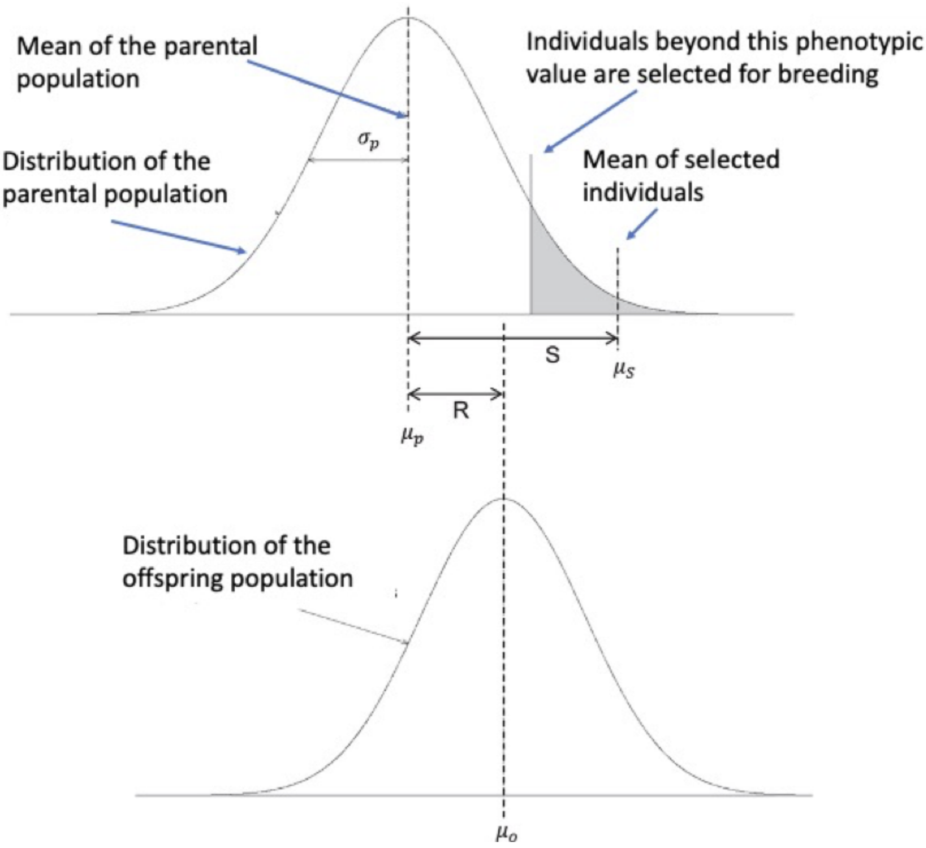


Figure 1: Schematic of the Breeder's Equation showing the relationship between selection differential ($S$) and response to selection ($R$). The selected parents have a mean phenotype greater than the population mean; the offspring generation shifts toward the selected parents but regresses toward the population mean by a factor determined by heritability ($h^2$).

### 6.4 A Worked Example

**Trait**: Body weight in mice

| Quantity | Value |
|---|---|
| Population mean weight ($\bar{P}_{\text{pop}}$) | 20 g |
| Mean weight of selected parents ($\bar{P}_{\text{selected}}$) | 24 g |
| Selection differential ($S$) | $24 - 20 = 4$ g |
| Narrow-sense heritability ($h^2$) | 0.4 |

**Predicted response**:

$$R = 0.4 \times 4 = 1.6 \text{ g}$$

**Predicted offspring mean**:

$$\bar{P}_{\text{offspring}} = 20 + 1.6 = 21.6 \text{ g}$$

**Interpretation**: Even though selected parents were 4 g heavier than average, offspring are only 1.6 g heavier. The "missing" 2.4 g is due to non-additive effects ($D, I$) and residual effects ($E$).

## 6.5 Real-World Example: Corn Selection Experiment

**Trait**: Oil content in corn (maize)

| **Generation** | **Population mean oil content** |
|---|---|
| Starting population | 5% |
| Selected parents | 7% ($S = 2\%$) |
| $h^2$ estimated | 0.7 |

**Predicted response**:

$$R = 0.7 \times 2\% = 1.4\%$$

**Predicted offspring mean**:

$$5\% + 1.4\% = 6.4\%$$

**What actually happened**: First generation response matched prediction closely. After many generations, response slowed (due to depletion of additive genetic variance, fixation of alleles, or negative pleiotropy).

> **Important lesson**: The Breeder's Equation predicts response for **one generation**. Over multiple generations, $h^2$ can change as allele frequencies shift and genetic variance is exhausted.

## 6.6 Limitations and Assumptions of the Breeder's Equation

| **Assumption** | **What it means** | **What violates it?** |
|---|---|---|
| $h^2$ known and constant | Heritability from population applies to selected group | Selection changes allele frequencies |
| No G×E interaction | Genetic effects consistent across environments | Different environments for parents vs. offspring |
| Random mating | Selected parents mate randomly | Assortative mating or inbreeding |
| No common environment | Parents and offspring in comparable environments | Parental effects (e.g., maternal nutrition) |
| Large population | No random drift | Small populations |
| No correlated selection | Selecting on one trait doesn't affect others | Genetic correlations with other traits |

> **Critical point**: The Breeder's Equation is a **one-generation prediction tool**. For long-term selection, re-estimate $h^2$ periodically.

## 6.7 Why the Breeder's Equation Is So Important

| Field | Application |
|---|---|
| Agriculture | Predict improvement in milk yield, growth rate, crop yield per generation |
| Evolutionary biology | Predict speed of adaptation to climate change, predation, new food sources |
| Conservation | Predict whether a small population can recover from inbreeding depression |
| Human genetics | Understand why offspring of tall parents are tall, but shorter (regression to the mean) |

> **The big idea**: The Breeder's Equation separates **what you can achieve** ($S$) from **what you can transmit** ($h^2$). You can select hard (large $S$), but if $h^2$ is small, response ($R$) will be small.

## 6.8 Summary of Step VI

| Concept | Definition |
|---|---|
| Breeder's Equation | $R = h^2 \times S$ |
| Selection differential ($S$) | Mean of selected parents  mean of population |
| Response ($R$) | Mean of offspring  mean of population |
| Heritability ($h^2$) | Proportion of phenotypic variance that is additive genetic |
| Key insight | Only the additive part of selection ($h^2 \times S$) is transmitted |

**Take-home message**:

> *The Breeder's Equation is the central predictive tool of quantitative genetics. It tells us that response to selection depends on two things: how extreme the selected parents are ($S$) and how much of that extremity is due to additive genetic effects ($h^2$). If you want rapid improvement, you need both strong selection (large $S$) and high heritability (large $h^2$).*

## 6.9 The Complete Journey: From Individual to Prediction

| Step | Question | Answer |
|---|---|---|
| Step II | What makes up an individual's phenotype? | $P = A + D + I + E$ |
| Step III | What is passed to offspring? | Only $A$ (breeding value) |
| Step IV | How do we measure $A$? | Progeny testing: $A = 2 \times (M - \bar{P})$ |
| Step V | How do we estimate $h^2$? | Mid-parent offspring regression (slope = $h^2$) |
| Step VI | How do we predict response? | $R = h^2 \times S$ |

## 6.10 A Note on Precision: Two Concepts of Heritability

Throughout this module, we have used $h^2$ to denote narrow-sense heritability and stated that:

1. It can be estimated as the slope of the parent-offspring regression ($b_{OP}$)
2. It appears in the Breeder's Equation as $R = h^2 \times S$

These two statements are often presented as if they are saying the same thing. Strictly speaking, they refer to **two conceptually distinct quantities** that happen to be numerically equal under a specific set of assumptions.

**The Distinction**:

| Concept | Definition | Context |
|---|---|---|
| Fisherian $h^2$ | Slope of parent-offspring regression ($b_{OP}$) | Descriptive statistic; answers: "Given current population, what is expected phenotypic resemblance?" |
| Breeder's Equation $h^2$ | Ratio $V_A/V_P$ | Predictive parameter; answers: "If I apply selection differential $S$, what will be response $R$?" |

**When are they equal?** Under ideal conditions:

- Random mating
- No selection on the parents used for estimation
- No common environment
- No G×E interaction
- No maternal effects
- Large population (no drift)

Then $b_{OP} = V_A/V_P = h^2$.

**When can they diverge?**

| Violation | Effect on $b_{OP}$ | Effect on $V_A/V_P$ |
|---|---|---|
| Assortative mating | Increases (inflated resemblance) | Unchanged (but selection response affected) |
| Selection on parents | Biases downward if selected parents used | Not applicable |
| Common environment | Increases (shared environment inflates) | Unchanged |
| Maternal effects | Increases | Unchanged |

**Why this matters for the Breeder's Equation**: The Breeder's Equation requires $h^2 = V_A/V_P$ as the correct scaling factor. If you use an observed $b_{OP}$ inflated by assortative mating or common environment, you will **overpredict** response. If you use $b_{OP}$ estimated from a population under selection, you may **underpredict** response.

**Practical take-home**:

> *The slope of the parent-offspring regression is an* ***estimate*** *of $h^2$* ***only under ideal conditions****. In real populations, always consider whether assumptions (random mating, no common environment, etc.) hold. If they do not, the regression slope may not equal $V_A/V_P$, and using it in the Breeder's Equation will give incorrect predictions.*

**For the advanced student**: The rigorous approach is:

1. Define $h^2$ as $V_A/V_P$.
2. Estimate $V_A$ and $V_P$ using methods that account for violations (e.g., animal models, mixed models).
3. Use that $h^2$ in the Breeder's Equation.
4. Recognize that parent-offspring regression is a special case estimator valid only under specific assumptions.

For an introductory course, the simplification that "regression slope $= h^2$" is widely used and acceptable, provided students are aware that it rests on assumptions that may not always hold.

# 7 Glossary of Terms

| Term | Symbol | Definition |
|---|---|---|
| Additive genetic value | $A$ | Sum of the average effects of all alleles an individual carries. Also called breeding value. |
| Breeder's Equation | $R = h^2 S$ | Fundamental equation predicting response to selection. |
| Breeding value | $A$ | Twice the deviation of an individual's progeny mean from the population mean (when mated randomly). The part of phenotype transmitted to offspring. |
| Dominance deviation | $D$ | Deviation from additivity due to interaction between two alleles at the same locus. |
| Epistatic deviation | $I$ | Deviation from additivity due to interactions between alleles at different loci. |
| Heritability (broad-sense) | $H^2$ | Proportion of phenotypic variance due to all genetic effects: $H^2 = V_G/V_P$. |
| Heritability (narrow-sense) | $h^2$ | Proportion of phenotypic variance due to additive genetic effects: $h^2 = V_A/V_P$. Predicts response to selection. |
| Mid-parent | - | The average of the two parents' phenotypic values for a trait. |
| Phenotype | $P$ | The observed trait value of an individual. |
| Population mean | $\bar{P}$ | The average phenotypic value of all individuals in a population. |
| Regression to the mean | - | Tendency for extreme parents to produce offspring closer to the population mean. Degree determined by $h^2$. |
| Residual deviation | $E$ | Unexplained variance in phenotype after accounting for genetic components. Includes environment, measurement error, developmental noise. |
| Response to selection | $R$ | Difference between mean of offspring generation and mean of original population. |
| Selection differential | $S$ | Difference between mean of selected parents and mean of original population. |
| Variance (additive genetic) | $V_A$ | Variance of breeding values in a population. Component that responds to selection. |
| Variance (dominance) | $V_D$ | Variance due to dominance deviations. Not transmitted. |
| Variance (epistatic) | $V_I$ | Variance due to epistatic deviations. Not reliably transmitted. |
| Variance (phenotypic) | $V_P$ | Total variance of a trait in a population. $V_P = V_A + V_D + V_I + V_E$ (under simplifying assumptions). |
| Variance (residual) | $V_E$ | Variance due to residual (non-genetic) factors. |

# 8 Summary Tables

## 8.1 Table 1: The Core Model

| **Component** | **Symbol** | **Transmitted to offspring?** | **Role in Breeder's Equation** |
|---|---|---|---|
| Additive | $A$ | Yes | The only component that contributes to $R$ |
| Dominance | $D$ | No | Not transmitted |
| Epistasis | $I$ | No | Not transmitted |
| Residual | $E$ | No | Not transmitted |

## 8.2 Table 2: Key Equations

| **Equation** | **Name** | **Purpose** |
|---|---|---|
| $P = A + D + I + E$ | Phenotypic decomposition (individual) | Describes what makes up an individual's phenotype |
| $V_P = V_A + V_D + V_I + V_E$ | Variance partition (population) | Partitions phenotypic variance into causal components |
| $A = 2 \times (M - \bar{P})$ | Breeding value definition | Defines breeding value operationally |
| Slope $= h^2$ | Mid-parent regression | Estimates narrow-sense heritability |
| $A = h^2 \times (P - \bar{P})$ | Breeding value prediction | Predicts breeding value from phenotype |
| $R = h^2 \times S$ | Breeder's Equation | Predicts response to selection |

## 8.3 Table 3: Heritability Values and Their Interpretation

| **$h^2$ value** | **Interpretation** | **Example traits** |
|---|---|---|
| $h^2 \approx 0$ | Almost no additive genetic variance; response negligible | Some fitness-related traits in wild populations |
| $0.2 < h^2 < 0.4$ | Moderate heritability; slow but measurable response | Human IQ (estimated), many life-history traits |
| $0.4 < h^2 < 0.6$ | High heritability; rapid response | Height in many populations, milk yield in dairy cattle |
| $h^2 \approx 1$ | All variation additive genetic; offspring exactly resemble parents (minus environment) | Rare – requires no dominance, epistasis, or residual variance |

### 8.4 Table 4: Historical Timeline

| Year | Event | Key figures |
|---|---|---|
| 1865 | Mendel's pea experiments | Gregor Mendel |
| 1890–1910 | Mendelian–Biometrician debate | Bateson, de Vries vs. Galton, Pearson |
| 1918 | Fisher's paper resolves debate; foundation of QG | R.A. Fisher |
| 1920s | Development of ANOVA | R.A. Fisher |
| 1930s–40s | Modern Evolutionary Synthesis | Fisher, Haldane, Wright |
| 1940s–50s | Epistasis added to QG model | Kenneth Mather et al. |

## 9 Final Take-Home Messages

1. **Most traits of biological and economic importance are continuous**. You cannot use simple Mendelian ratios to predict inheritance of height, yield, or disease risk.

2. **Fisher's great synthesis (1918)** showed that continuous variation arises from many Mendelian loci of small effect, plus residual variance. This resolved the Mendelian–Biometrician debate and founded quantitative genetics.

3. **Only the additive component ($A$) of an individual's phenotype is transmitted to offspring.** Dominance ($D$), epistasis ($I$), and residual variance ($E$) are not heritable.

4. **Breeding value** is the operational measure of what an individual passes on. It is defined as twice the deviation of an individual's progeny mean from the population mean.

5. **Narrow-sense heritability ($h^2$)** is the proportion of phenotypic variance due to additive genetic variance. It predicts response to selection.

6. **Mid-parent offspring regression** is the classic method for estimating $h^2$. The slope of the regression line equals $h^2$ under ideal conditions.

7. **The Breeder's Equation ($R = h^2 S$)** is the central predictive tool of QG. Response to selection depends on both the intensity of selection ($S$) and heritability ($h^2$).

8. **All QG predictions are population-level and probabilistic.** You cannot predict individual offspring outcomes, but you can predict the mean of the offspring generation.

9. **Heritability and breeding value are not fixed properties.** They depend on the population, environment, and mating system.

10. **The power of QG is that it works without knowing the underlying genes.** This is why it remains essential for breeding, evolution, and complex trait analysis even in the genomic era.

11. **Precision note**: The $h^2$ estimated from parent-offspring regression and the $h^2$ used in the Breeder's Equation are conceptually distinct, though numerically equal under ideal conditions (random mating, no selection on estimation sample, no common environment, etc.). Violations can cause divergence, so always interpret heritability estimates with caution.

# ESSENTIALS VERSION

*Approximately 4 pages. Core logical flow, key equations, and main takeaways.*

# Quantitative Genetics: Essentials

## What Is Quantitative Genetics?

Quantitative Genetics (QG) deals with **continuously varying traits** (height, yield, cholesterol) influenced by many genes of small effect plus environment. It predicts **population responses** to selection, not individual outcomes.

**The problem QG solves**: Mendelian genetics explains discrete traits (pea color) but not continuous ones. Fisher (1918) showed that **many Mendelian loci + residual variance** produce continuous variation.

## The Core Model: What Makes Up a Phenotype?

$$P = A + D + I + E$$

| Symbol | Meaning | Passed to offspring? |
|---|---|---|
| $P$ | Phenotypic value (measured trait) | — |
| $A$ | Additive genetic value (breeding value) | **Yes** |
| $D$ | Dominance deviation (within-locus interaction) | No |
| $I$ | Epistatic deviation (across-locus interaction) | No |
| $E$ | Residual deviation (unexplained variance) | No |

> **The central insight**: Only the additive component ($A$) transmits from parent to offspring. Dominance, epistasis, and residual effects are not inherited.

**Why only $A$?** Parents pass **alleles**, not genotypes. Dominance requires a specific pair of alleles (broken by meiosis). Epistasis requires specific multi-locus combinations (scrambled by recombination). $E$ is not encoded in DNA.

## From Individuals to Populations: Variance Partition

$$V_P = V_A + V_D + V_I + V_E$$

Narrow-sense heritability:

$$h^2 = \frac{V_A}{V_P}$$

## Breeding Value ($A$)

**Definition**: Twice the deviation of an individual's progeny mean from the population mean (when mated randomly).

$$A = 2 \times (M - \bar{P})$$

- $M$ = mean of progeny

- $\bar{P}$ = population mean

**Interpretation**: An individual with breeding value $+x$ produces offspring that exceed the population mean by $x/2$ (because it contributes half the alleles).

**Important properties**:

- Breeding value is **population-dependent** (changes with allele frequencies, environment, mating system)
- Variance of breeding values = $V_A$ (additive genetic variance)

### Narrow-Sense Heritability ($h^2$)

$$h^2 = \frac{V_A}{V_P}$$

**Interpretation**: Proportion of phenotypic variance due to additive genetic effects. Predicts response to selection.

**How to estimate** $h^2$ (mid-parent offspring regression):

- Regress offspring phenotype on mid-parent (average of two parents)
- **Slope** = $h^2$ (under ideal conditions: random mating, no common environment, no selection on parents used for estimation)

**Predict breeding value from phenotype**:

$$A = h^2 \times (P - \bar{P})$$

### The Breeder's Equation

$$\boxed{R = h^2 \times S}$$

| Symbol | Name | Definition |
|---|---|---|
| $R$ | Response | Mean of offspring  mean of original population |
| $h^2$ | Heritability | $V_A/V_P$ |
| $S$ | Selection differential | Mean of selected parents  mean of original population |

**What it tells us:**

- If $h^2 = 1$, offspring shift as much as selected parents ($R = S$)
- If $h^2 = 0$, no response regardless of selection ($R = 0$)
- If $0 < h^2 < 1$, offspring shift less than parents ($R < S$)

**Why** $h^2$ **discounts** $S$: Only the additive part of the parental phenotype is transmitted. Non-additive parts ($D, I, E$) are not inherited.

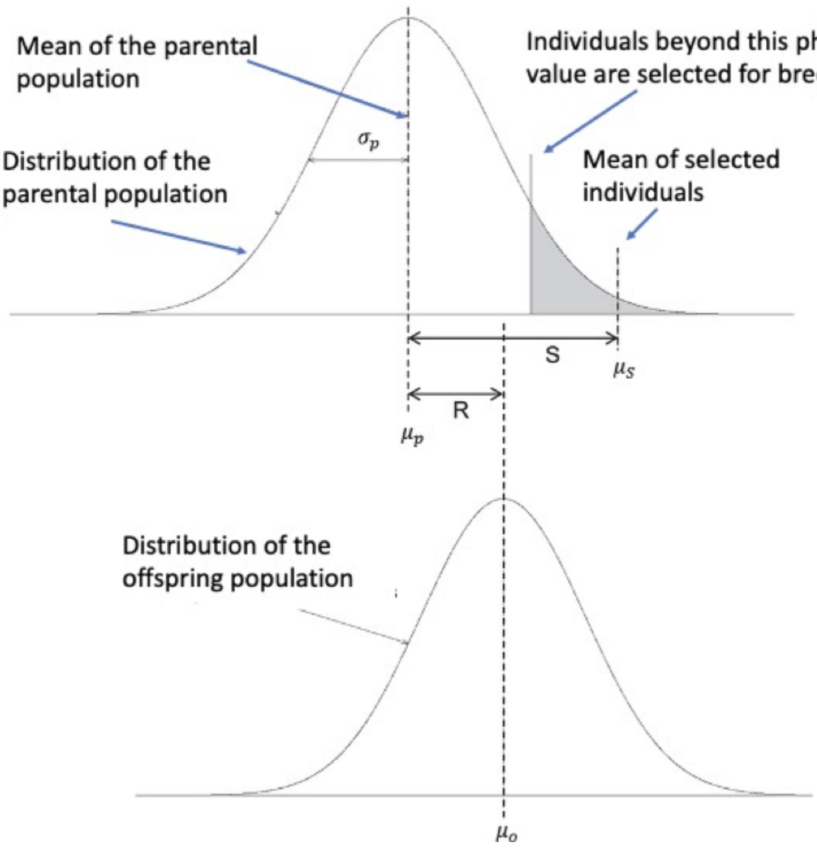


Figure 2: Schematic of the Breeder's Equation showing the relationship between selection differential ($S$) and response to selection ($R$).

### One Worked Example

**Trait**: Body weight in mice

- Population mean = 20 g
- Selected parents mean = 24 g → $S = 4$ g
- $h^2 = 0.4$

**Prediction**:

$$R = 0.4 \times 4 = 1.6 \text{ g}$$

Offspring mean = $20 + 1.6 = 21.6$ g

### Key Assumptions (for the simple formulas to hold)

| Assumption | Why it matters |
|---|---|
| Random mating | Affects variance of mid-parent values |
| No common environment | Parent-offspring resemblance not inflated by shared conditions |
| No G×E | Genetic effects consistent across environments |
| No selection on parents used for estimation | Otherwise regression slope biased |
| Large population | No random drift |

**Warning**: If assumptions are violated, estimates may be biased. Heritability and breeding value are **population-specific**, not universal constants.

### The Complete Logical Flow

| Step | Question | Answer |
|---|---|---|
| 1 | What makes up a phenotype? | $P = A + D + I + E$ |
| 2 | How is variance partitioned? | $V_P = V_A + V_D + V_I + V_E$ |
| 3 | What is passed to offspring? | Only $A$ (breeding value) |
| 4 | How to measure $A$? | Progeny testing: $A = 2(M - \bar{P})$ |
| 5 | How to estimate $h^2$? | Mid-parent regression slope = $h^2$ |
| 6 | How to predict response? | $R = h^2 \times S$ |

### Precision Note: Two Concepts of Heritability

The $h^2$ estimated from parent-offspring regression and the $h^2$ used in the Breeder's Equation are **conceptually distinct** but **numerically equal under ideal conditions** (random mating, no selection on estimation sample, no common environment, no G×E).

If assumptions are violated (e.g., assortative mating), the regression slope may not equal $V_A/V_P$, and using it in the Breeder's Equation will mispredict response.

**Take-home**: The regression slope is an **estimate** of $h^2$ only under ideal conditions. Always check assumptions.

### Why QG Matters

| Field | Application |
|---|---|
| Agriculture | Predict improvement in yield, milk, growth rate |
| Evolutionary biology | Predict speed of adaptation |
| Conservation | Predict recovery from inbreeding |
| Human genetics | Understand regression to the mean (tall parents have tall but shorter offspring) |

### Final Take-Home

**Quantitative Genetics allows prediction of population response to selection without knowing individual genes. The Breeder's Equation ($R = h^2 S$) separates what you can achieve (selection differential) from what you can transmit (heritability). Only additive genetic variance drives response.**

*For detailed derivations, examples, historical background, glossary, and full tables, see the detailed version of this module.*